CHAPTER 13

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Resistance Training Concepts

OBJECTIVES

After studying this chapter, you will be able to:

- ✔ Describe the stages of the general adaptation syndrome.
- ✔ Define and describe the principle of adaptation and specificity.
- ✔ Define stability, muscular endurance, muscular hypertrophy, strength, and power.
- ✔ List and define the various stages of strength and training systems.

Introduction to Resistance Training

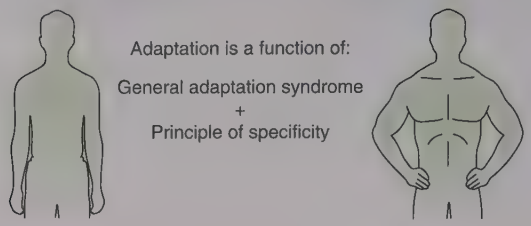
The final step in completing the OPT™ template is planning the resistance training component of an exercise program. The resistance training section of the OPT template requires filling in the appropriate exercises for each body part (chest, back, shoulders, legs, etc.), the sets, repetitions, tempo (speed of repetition), and rest interval (amount of rest between each exercise) based on assessment outcomes and goals for each client. Personal trainers need to understand the science and principles of resistance training to be effective at designing individually tailored safe and appropriate resistance training workouts. This chapter explores some of the most important concepts involved in designing and implementing resistance training programs for a variety of clients. The focus of this chapter includes principles of adaptation, progressive strength adaptations derived from resistance training, training systems used to acquire strength, and specific resistance training exercise progressions.

Principle of Adaptation

One of the many unique qualities of the human body is its ability to adapt or adjust its functional capacity to meet the desired needs **Figure 13.1** (1–6). The ability of the human body to respond and adapt to an exercise stimulus is perhaps one of the

FIGURE 13.1

Resistance training principle of adaptation.



most important concepts of training and conditioning. Some form of adaptation is the primary goal of most exercise training programs. Whether the goal is cosmetic in nature or health- or performance-related, resistance training has been shown to produce a variety of desirable effects Table 13.1.

General Adaptation Syndrome

General adaptation syndrome a term used to describe how the body responds and adapts to stress.

The optimal state for the human movement system to be in is one of physiologic balance or homeostasis. The **general adaptation syndrome** (GAS) is a term used to describe how the body responds and adapts to stress. In this case the stress being placed on the body is the weight being lifted during resistance training. This general pattern of adaptation was first described by Hans Selye, a Canadian physician, who stated that exercise, including resistance training, can be considered a good form of stress called "eustress" that over time allows the human movement system to adapt and thus be able to maintain homeostatic states under a variety of conditions. For adaptation to occur, the body must be confronted with a stressor or some form of stress that creates the need for a response Table 13.2 (6). Selye outlined three stages of response to stress (6):

- Alarm reaction
- Resistance development
- Exhaustion

TABLE 13.1 Adaptive Benefits of Resistance Training

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Physiologic	Physical	Performance
Improved cardiovascular efficiency	Increased tissue (muscle, tendons, ligaments) tensile strength	Increased neuromuscular control (coordination)
Beneficial endocrine (hormone) and serum lipid (cholesterol) adaptations	Increased cross-sectional area of muscle fibers	Increased endurance
Increased bone density	Decreased body fat	Increased strength
Increased metabolic efficiency (metabolism)		Increased power

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TABLE 13.2 The General Adaptation Syndrome

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Stage	Reaction
Alarm reaction	Initial reaction to stressor such as increased oxygen and blood supply to the necessary areas of the body
Resistance development	Increased functional capacity to adapt to stressor such as increasing motor unit recruitment
Exhaustion	A prolonged intolerable stressor produces fatigue and leads to a breakdown in the system or injury

Alarm Reaction Stage

The **alarm reaction** is the initial reaction to a stressor. The alarm reaction activates a number of physiological and psychological protective processes within the body. During the initial sessions of resistance training programs, the body is forced to try and adapt to increased amounts of force on bones, joints, muscles, connective tissues, and the nervous system. During the alarm stage of resistance training, numerous physiologic responses occur, including an increase in oxygen and blood supply as well as neural recruitment to the working muscles. Initially, an individual's body may be very inefficient at responding to the demands placed on it during resistance training. But gradually over time and by applying the principle of progressive overload, the body increases its ability to meet the demands being placed on it (6).

Alarm reaction The alarm reaction is the initial reaction to a stressor.

Delayed-onset muscle soreness Pain or discomfort often felt 24 to 72 hours after intense exercise or unaccustomed physical activity.

Consider the typical response to either unaccustomed exercise or a sudden increase in a training program. The new work is performed, and during the next 2 to 3 days, the muscles may exhibit classic **delayed-onset muscle soreness** or DOMS (7). During this period of DOMS, any attempt at replicating or advancing the soreness-inducing exercise will be limited by the factors contributing to the soreness. This could be considered an "alarm reaction." Most experts agree minimizing DOMS involves starting a progressive training program at a low intensity and introducing overload gradually (7).

Resistance Development Stage

During the **resistance development** stage, the body increases its functional capacity to adapt to the stressor. After repeated training sessions, the human movement system will increase its capability to efficiently recruit muscle fibers and distribute oxygen and blood to the proper areas in the body. Once adaptation has occurred, the body will require increased stress or overload to produce a new response and a higher level of fitness (6).

Resistance development The body increases its functional capacity to adapt to the stressor.

Personal trainers often understand this adaptation response, but use it improperly by *only* manipulating the amount of weight the client uses when, in fact, this is but one of many ways to increase stress on the body. Chapter fourteen will discuss the importance of manipulating the many acute variables (sets, reps, intensity, rest periods, exercise selection, etc.) for optimal adaptation while avoiding breakdown or exhaustion.

In the example of unaccustomed exercise, once the DOMS subsides, further work will be met with less and less soreness so that performance may gradually advance. This would be "resistance development." Performance will continue to improve until some new performance plateau is reached and will be maintained if training is maintained.

Exhaustion Stage

Exhaustion Prolonged stress or stress that is intolerable and will produce exhaustion or distress to the system.

Prolonged stress or intolerable amounts of stress can lead to **exhaustion** or distress. When a stressor is too much for any one of the physiologic systems to handle, it causes a breakdown or injury such as (6):

- Stress fractures
- Muscle strains
- Joint pain
- Emotional fatigue

In turn, many of these types of injuries can lead to the initiation of the cumulative injury cycle.

Avoiding the pitfalls of the exhaustion stage is one of the main reasons for using the OPT model (a systematic, progressive training program) that is based on science and proven application. Resistance training, as well as other forms of training, must be cycled through different stages that increase stress placed on the human movement system, but also allow for sufficient rest and recuperation periods. The term used to describe this approach, in which a training program is divided into smaller, progressive stages, is **periodization**. Additional information on periodization of training (OPT model) is detailed in chapter 14.

Periodization Division of a training program into smaller, progressive stages.

In the above example, if the resistance is continually increased with the intention of stressing specific muscles or muscle groups to produce an increase in size or strength, it can lead to injury of the muscle, joint, or connective tissue, especially if the resistance is added too quickly or inadequate rest and recovery periods are not planned for. Training-related injuries occur more often to connective tissue (such as ligaments and tendons) than muscles because connective tissues lack blood supply (8). Different tissues in the body (muscle fibers versus connective tissue) each have their own adaptive potential to stresses. Thus, training programs should provide a variety of intensities and stresses to optimize the adaptation of each tissue to ensure the best possible results. Adaptation can be more specifically applied to certain aspects of the human movement system depending on the training technique(s) used, which is the basis of the principle of specificity.

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Did You Know?

Overtraining syndrome commonly occurs in athletes or fitness enthusiasts who are training beyond the body's ability to recover. When an individual is performing excessive amounts of exercise without proper rest and recovery, there may be some harmful side effects. Some of these side effects may include decreased performance, fatigue, altered hormonal states, poor sleeping patterns, reproductive disorders, decreased immunity, loss of appetite, and mood disturbances (9).

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The Principle of Specificity: The SAID Principle

The **principle of specificity**, often referred as the SAID (**specific adaptation to imposed demands**) principle, states that the body will specifically adapt to the type of demand placed on it. For example, if someone repeatedly lifts heavy weights, that person will produce higher levels of maximal strength. Conversely, if a person repeatedly lifts lighter weights for many repetitions, that person will develop higher levels of muscular endurance.

Principle of specificity *or* specific adaptation to imposed demands (SAID principle) Principle that states the body will adapt to the specific demands that are placed on it.

According to the principle of specificity, training programs should reflect the desired outcome(s). When applying the SAID principle to any training program, it is important to remember that the body is made up of many types of tissues, and these tissues may respond differently to the same stimulus. To make the principle of specificity a safe and effective tool, it must be used appropriately.

Remember that type I muscle fibers function differently than type II muscle fibers. Type I or slow-twitch muscle fibers are smaller in diameter, slower to produce maximal tension, and more resistant to fatigue. Type I fibers are important for muscles that need to produce long-term contractions necessary for stabilization, endurance, and postural control. Type II or fast-twitch muscle fibers are larger in size, quick to produce maximal tension, and fatigue more quickly than type I fibers. These fibers are important for muscles producing movements requiring force and power such as performing a sprint. To train with higher intensities, proper postural stabilization is required. Therefore, tissues need to be trained differently to prepare them for higher levels of training. This is the specific purpose behind periodization and the OPT model **Figure 13.2**.

The degree of adaptation that occurs during training is directly related to the mechanical, neuromuscular, and metabolic specificity of the training program (3,5). To effectively achieve program goals for clients, personal trainers need to consistently evaluate the need to manipulate the exercise routine to meet actual training goals. Remember, the body can only adapt if it has a reason to adapt.



FIGURE 13.2
The OPT model.

Mechanical specificity Refers to the weight and movements placed on the body.

Neuromuscular specificity Refers to the speed of contraction and exercise selection.

Metabolic specificity Refers to the energy demand placed on the body.

- **Mechanical specificity** refers to the weight and movements placed on the body (3,5). To develop muscular endurance of the legs requires light weights and high repetitions when performing leg-related exercises. To develop maximal strength in the chest, heavy weights must be used during chest-related exercises.
- **Neuromuscular specificity** refers to the speed of contraction and exercise selection (10–12). To develop higher levels of stability while pushing, chest exercises will need to be performed with controlled, unstable exercises, at slower speeds **Figure 13.3**. To develop higher levels of strength, exercises should be performed in more stable environments with heavier loads to place more of an emphasis on the prime movers **Figure 13.4**. To develop higher levels of power, low-weight, high-velocity contractions must be performed in a plyometric manner **Figure 13.5**.
- **Metabolic specificity** refers to the energy demand placed on the body. To develop endurance, training will require prolonged bouts of exercise, with minimal rest periods between sets. Endurance training primarily uses aerobic pathways to supply energy for the body. To develop maximal strength or power, training will require longer rest periods, so the intensity of each bout of exercise remains high. Energy will be supplied primarily via the anaerobic pathways (13,14).

FIGURE 13.3

Training for stability.



FIGURE 13.4

Training for strength.



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FIGURE 13.5

Training for power.

It is important for personal trainers to remember that a client's training program should be designed to meet the specific demands of their daily life and health and wellness goals. The following example applies the concept of specificity to a client whose goal is body fat reduction (15–18).

1. Mechanically, the body burns more calories when movements are performed while standing (versus a seated or lying position) and using moderate weights. An example would be performing standing cable rows versus seated machine rows.
2. From a neuromuscular standpoint, the body burns more calories when more muscles are being used for longer periods in controlled, unstable environments. An example would be performing a single-leg dumbbell shoulder press versus a seated machine shoulder press.
3. Metabolically, the body burns more calories when rest periods are short to minimize full recuperation. An example would be to have your clients perform resistance training exercises in a circuit fashion with no rests between sets.

Applying the principles of specificity to a training program for weight loss, the client should perform the majority of exercises while standing and using moderate weights. The client should also recruit and use as many muscles as possible during each exercise and carefully monitor rest periods for greater caloric expenditure.

SUMMARY

A well-designed, integrated training program produces optimal levels of:

- Flexibility
- Endurance
- Neuromuscular control
- Alterations in body composition
- Strength
- Power

To achieve optimal training results, the body must adapt to specifically imposed demands and stresses. The ability to adapt to stress is known as the general adaptation syndrome. There are three stages of response to stress: (1a) alarm reaction (or initial activation of protective processes within the body), (b) resistance development (or an

increase in the functional capacity to adapt to a stressor), and (c) exhaustion (or stress that is too much for the system and causes an injury). To avoid injury, adaptive programs must include carefully and methodically planned (periodization) cycles through different stages, which allows for sufficient rest and recuperation. Adaptive programs have several benefits, including:

- Physiologic
  - Improved cardiovascular efficiency
  - Beneficial endocrine (hormone) and serum lipid (cholesterol) adaptations
  - Increased bone density
  - Increased metabolic efficiency (metabolism)
- Physical
  - Increased tissue (muscles, tendons, ligaments) tensile strength
  - Increased cross-sectional area of muscle fibers
  - Decreased body fat
- Performance
  - Increased neuromuscular control (coordination)
  - Increased endurance
  - Increased strength
  - Increased power

The type of training adaptation and the goals of the client should determine (or dictate) the design of the training program. There are different types of strength and different systems of strength training that can be used to create a more individualized and systematic program for clients. In addition, different tissues in the body each respond to different stresses as seen in the principle of specificity or specific adaptation to imposed demands (or SAID principle). Training programs should provide a variety of intensities and stresses to optimize the adaptation of each tissue to ensure the best possible results. The degree of adaptation that occurs during training is directly related to the mechanical, neuromuscular, and metabolic specificity of the training program.

Progressive Adaptations from Resistance Training

The concept of adaptation makes it clear that some type of change will occur based on the stresses placed on the body. Resistance training programs are designed to produce changes that result in various adaptations. Whether the goal is to increase muscle endurance, strength, hypertrophy, or power or to reduce body fat and improve overall health, the use of resistance training is an important component of any fitness program. This will help ensure optimal health and longevity for the client. As clients develop greater strength and endurance, they can train for longer periods before reaching the exhaustion stage (general adaptation syndrome), which leads to greater degrees of change and adaptation realized over time. The main adaptations that occur from resistance training include stabilization, muscular endurance, hypertrophy, strength, and power.

Stabilization

Stabilization is the human movement system's ability to provide optimal dynamic joint support to maintain correct posture during all movements. In other words, stabilization

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is getting the right muscles to fire, with the right amount of force, in the proper plane of motion, and at the right time. This requires high levels of muscular endurance for optimal recruitment of prime movers to increase concentric force production and reduce eccentric force. Repeatedly training with controlled, unstable exercises increases the body's ability to stabilize and balance itself (19–21). Conversely, if training is not performed with controlled unstable exercises, clients will not gain the same level of stability and may even worsen (21–23). Research shows that improper stabilization can negatively affect a muscle's force production (24). Stability is an important training adaptation because it increases the ability of the kinetic chain to stabilize the lumbo-pelvic-hip complex and joints during movement to allow the arms and legs to work more efficiently.

Muscular Endurance

Muscular endurance is the ability to produce and maintain force production for prolonged periods of time. Improving muscular endurance is an integral component of all fitness programs. Developing muscular endurance helps to increase core and joint stabilization, which is the foundation on which hypertrophy, strength, and power are built. Training for muscular endurance of the core focuses on the recruitment of muscles responsible for postural stability, namely, type I muscle fibers.

Muscular endurance The ability to produce and maintain force production for prolonged periods of time.

Research has shown that resistance training protocols using high repetitions are the most effective way to improve muscular endurance (25–27). In addition, a periodization training program can enhance local muscular endurance as well (26,28,29), and after an initial training effect in previously untrained individuals, multiple sets of periodized training may prove superior to single-set training for improving muscle endurance (26). Campos et al. (25) found that higher repetitions (2 sets of 20–28 repetitions with 1-minute rest periods beginning at 2 days/week) increased local muscle endurance and hypertrophy in untrained men after an 8-week training program, whereas Marx et al. (26) found that after the initial 12 weeks of training, multiple sets of up to 15 repetitions, 4 times a week for 6 months, resulted in a threefold decrease in body fat and an increase in local muscle endurance, as well as a significant increase in lean body mass.

Muscular Hypertrophy

Muscular hypertrophy is the enlargement of skeletal muscle fibers in response to being recruited to develop increased levels of tension, as seen in resistance training (30). Muscle hypertrophy is characterized by an increase in the cross-sectional area of individual muscle fibers resulting from an increase in myofibril proteins (myofilaments). Although the visible signs of hypertrophy may not be apparent for many weeks (4 to 8 weeks) in an untrained client, the process begins in the early stages of training, regardless of the intensity of training used (31–33).

Muscular hypertrophy Enlargement of skeletal muscle fibers in response to overcoming force from high volumes of tension.

Resistance training protocols that use low to intermediate repetition ranges with progressive overload lead to muscular hypertrophy. Structured progressive resistance training programs using multiple sets help to increase muscular hypertrophy in both younger and older men and women alike (25,28,34–37). Kraemer et al. (4) demonstrated that 24 weeks of training 3 days per week with 3 sets of 8–12 repetitions per exercise improved muscle hypertrophy and body composition. Thus, progressive resistance training programs using moderate to low repetition protocols with progressively higher loads will result in increased hypertrophy in older adults and men and women.

Strength

Strength The ability of the neuromuscular system to produce internal tension to overcome an external load.

Strength is the ability of the neuromuscular system to produce internal tension (in the muscles and connective tissues that pull on the bones) to overcome an external force. Whether the external force demands the neuromuscular system to produce stability, endurance, maximal strength, or power, internal tension within the muscles is what leads to force production. The degree of internal tension produced is the result of *strength adaptations*. The specific form of strength or internal tension produced from training is based on the type and intensity of training used by the client (principle of specificity).

Resistance training programs have traditionally focused on developing maximal strength in individual muscles, emphasizing one plane of motion (typically the sagittal plane) (3). Because all muscles function eccentrically, isometrically, and concentrically in all three planes of motion (sagittal, frontal, and transverse) at different speeds, training programs should be designed using a progressive approach that emphasizes the appropriate exercise selection, all muscle actions, and repetition tempos. This is discussed in greater detail in chapter 15.

Because muscle operates under the control of the central nervous system, strength needs to be thought of not as a function of muscle, but as a result of activating the neuromuscular system. Strength gains can occur rapidly in beginning clients and can increase with a structured, progressive resistance training program. One factor in increased strength is an increase in the number of motor units recruited, especially early in a training program (38–40). Using heavier loads increases the neural demand and recruitment of more muscle fibers until a recruitment plateau is reached, after which further increases in strength are a result of fiber hypertrophy (41–44).

Strength cannot be thought of in isolation. Strength is built on the foundation of stabilization requiring muscles, tendons, and ligaments to be prepared for the load that will be required to increase strength beyond the initial stages of training. Whereas stabilization training is designed with the characteristics of type I slow-twitch muscle fibers in mind (slow-contracting, low tension output, and resistant to fatigue), strength training is designed to match the characteristics of type II muscle fibers (quick-contracting, high tension output, prone to fatigue). Thus acute variables (sets, reps, intensities, etc.) are manipulated to take advantage of the specific characteristics of each fiber type. The majority of strength increases will occur during the first 12 weeks of resistance training from increased neural recruitment and muscle hypertrophy (26,29,34,45–48). Intermediate and advanced lifters will find it necessary to carry out a more demanding program in terms of training volume and intensity by following a sound periodized schedule.

Power

Power Ability of the neuromuscular system to produce the greatest force in the shortest time.

Power is the ability of the neuromuscular system to produce the greatest possible force in the shortest possible time. This is represented by the simple equation of force multiplied by velocity (49). Power adaptations build on stabilization and strength adaptations and then apply them at more realistic speeds and forces seen in everyday life and sporting activities. The focus of power-resistance training is getting the neuromuscular system to generate force as quickly as possible (rate of force production).

An increase in either force or velocity will produce an increase in power. Training for power can be achieved by increasing the weight (force), as seen in the strength adaptations, or increasing the speed with which weight is moved (velocity) (42,50–57). Power training allows for increased rate of force production by increasing the number

of motor units activated, the synchronization between them, and the speed at which they are activated (42,58,59). The general adaptation syndrome and principle of specificity both dictate that to maximize training for this type of adaptation, both heavy and light loads must be moved as fast as possible (in a controlled fashion). Thus, using both training methods in a superset fashion can create the necessary adaptations to enhance the body's ability to recruit a large number of motor units and increase the rate (speed) of activation (50–57). Early isokinetic work underscored the importance of speed of exercise, showing that training performed at high speeds led to better performance at the training speed and all movement speeds below the training speed.

SUMMARY

Resistance training programs produce physiologic changes that result in a variety of strength adaptations. Strength is the ability of the neuromuscular system to produce internal tension to overcome an external force. Traditionally, resistance training programs focused on developing maximal strength in individual muscles, whereas contemporary training programs emphasize appropriate exercise selection, all muscle actions, multiple planes of motion, and repetition tempos. Training adaptations can be divided into stabilization, muscular endurance, hypertrophy, strength, and power.

Stabilization adaptations should be the starting point for clients new to resistance training. Stabilization adaptations are best achieved using high repetitions with low to moderate volume and low to moderate intensity in a postural position that challenges the stability of the body.

Muscular endurance is the ability to produce and maintain force production for prolonged periods of time. Research has shown that resistance training protocols using high repetitions are the most effective way to improve muscular endurance.

Muscular hypertrophy is the enlargement of skeletal muscle fibers in response to increased volumes of tension resulting from resistance training. Resistance training protocols that use low to intermediate repetition ranges with progressive overload lead to muscular hypertrophy.

Strength is the ability of the neuromuscular system to produce internal tension (in the muscles and connective tissues that pull on the bones) to overcome an external force. The specific form of strength produced is based on the type and intensity of training used by the client (principle of specificity). Generally, strength adaptations should use low to moderate repetition routines with moderate to high volume and moderate to high intensity. Heavier weights and higher volumes of training are used to improve the function of motor units, while placing stress on the muscles to increase size or strength.

Power is the ability of the neuromuscular system to produce the greatest possible force in the shortest possible time. An increase in either force (weight) or velocity (speed with which weight is moved) will produce an increase in power. To maximize training for this adaptation, both heavy and light loads must be moved as fast and as controlled as possible.

Resistance Training Systems

Originally, power lifters, Olympic lifters, and bodybuilders designed most resistance training programs. Many of these styles of resistance training programs remain popular

today because of good marketing or "gym science," not because they have been proven to be scientifically superior to other forms of training programs that bring about increases in stabilization, strength, and power. Research has shown that following a systematic, integrated training program and manipulating key training variables achieve optimal gains in strength, neuromuscular efficiency, hypertrophy, and performance (4,27,28).

There are numerous training systems that can be used to structure resistance training programs for a variety of effects. Several of the most common training systems currently used in the fitness industry are presented in this chapter Table 13.3.

The Single-Set System

As the name suggests, the single-set system uses 1 set per exercise. It is usually recommended that single-set workouts be performed two times per week to promote sufficient development and maintenance of muscle mass (60). Although multiple-set training is promoted as being more beneficial for strength and hypertrophy gains in advanced clients, the single-set system has been shown to be just as beneficial for beginning-level clients (61–65).

Personal trainers are encouraged to explore the benefits and options of single-set workouts to further customize individual program design options. Single-set training systems are often negatively perceived for not providing enough stimuli for adaptation. However, when reviewing the physiology of how the human movement system operates, this notion may not be true (65). In fact, most beginning clients could follow a single-set program to allow for proper adaptive responses of the connective tissue and nervous system before engaging in more rigorous training systems. By encouraging clients to avoid lifting more than they can handle, synergistic dominance (synergists overcompensating for weak prime movers) and injury can be avoided.

TABLE 13.3 Resistance Training Systems

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Type	Definition
Single-set	Performing one set of each exercise
Multiple-set	Performing a multiple number of sets for each exercise
Pyramid	Increasing (or decreasing) weight with each set
Superset	Performing two exercises in rapid succession with minimal rest
Drop-sets	Performing a set to failure, then removing a small percentage of the load and continuing with the set
Circuit training	Performing a series of exercises, one after the other, with minimal rest
Peripheral heart action	A variation of circuit training that uses different exercises (upper and lower body) for each set through the circuit
Split-routine	A routine that trains different body parts on separate days
Vertical loading	Performing exercises on the OPT template one after the other, in a vertical manner down the template
Horizontal loading	Performing all sets of an exercise (or body part) before moving on to the next exercise (or body part)

The Multiple-Set System

The multiple-set system, on the other hand, consists of performing multiple numbers of sets for each exercise. The resistance (load), sets, and repetitions performed are selected according to the goals and needs of the client (66). Multiple-set training can be appropriate for both novice and advanced clients, but has been shown to be superior to single-set training for more advanced clients (29,66,67). The increased volume (sets, reps, and intensity) is necessary for further improvement, but must be administered appropriately to avoid overtraining (5).

The Pyramid System

The pyramid system involves a progressive or regressive step approach that either increases weight with each set or decreases weight with each set **Figure 13.6**. In the *light-to-heavy system*, the individual typically performs 10 to 12 repetitions with a light load and increases the resistance for each following set, until the individual can perform 1 to 2 repetitions, usually in 4 to 6 sets. This system can easily be used for workouts that involve only 2 to 4 sets or higher repetition schemes (12 to 20 repetitions). The *heavy-to-light system* works in the opposite direction, in which the individual begins with a heavy load (after a sufficient warm-up) for 1 to 2 repetitions, then decreases the load and increases the repetitions for 4 to 6 sets.

The Superset System

The superset system uses two exercises performed in rapid succession of one another. There are multiple variations of the superset system.

The first variation includes performing two exercises for the same muscle group back to back. For example, an individual may perform the bench press exercise immediately followed by push-ups to fatigue the chest musculature. Completing two exercises in this manner will improve muscular endurance and hypertrophy because the volume of work performed is relatively high. This style of supersets can use two, three (a tri-set), or more exercises (a giant set) for the target muscle group. The greater the number of exercises used, the greater the degree of fatigue experienced and demands on muscular endurance.

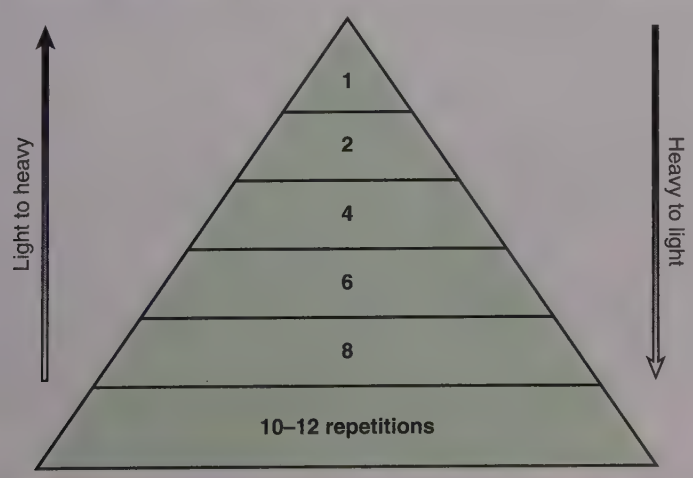


FIGURE 13.6
The pyramid system.

The second variation consists of performing two exercises back to back that involve antagonist muscle groups (e.g., chest and back or quadriceps and hamstring complex). Performing supersets in this manner allows a significant load to be placed on the target muscle during each set. This is possible because while the agonist is working, the antagonist is recovering, and vice versa.

Supersetting typically involves sets of 8 to 12 repetitions with no rest between sets or exercises; however, any number of repetitions can be used. The superset system is popular among bodybuilders and may be beneficial for muscular hypertrophy and muscular endurance.

Drop-Sets

A drop-set is a resistance training system that is popular among bodybuilders. It is a technique that allows a client to continue a set past the point at which it would usually terminate. Drop-sets involve performing a set to failure, then removing a small percentage of the load (5–20%) and continuing with the set, completing a small number of repetitions (usually 2–4). This procedure can be repeated several times (typically 2 to 3 drops per set). A set to failure followed by three successive load decrements performed with no rest would be referred to as a triple drop. Drop-sets are considered an advanced form of resistance training suitable for experienced lifters.

The Circuit-Training System

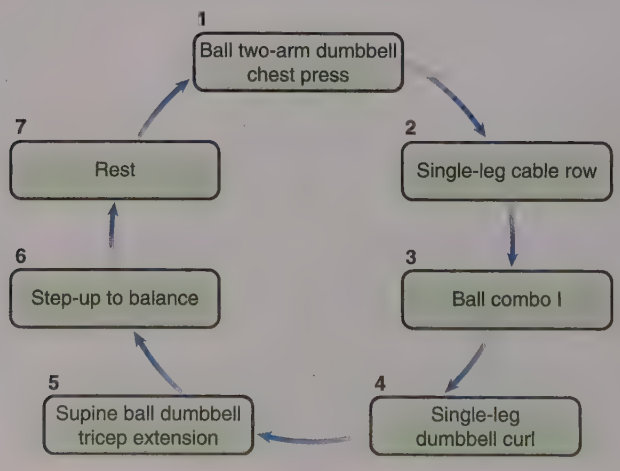
The circuit-training system consists of a series of exercises that an individual performs one after the other, with minimal rest between each exercise **Figure 13.7**. The typical acute variables for a circuit-training program include low to moderate number of sets (1–3), with moderate to high repetitions (8–20) and short rest periods (15–60 seconds) between exercises; however, these variables can be manipulated to enhance the desired effect (68,69). Circuit training is a great training system for individuals with limited time and for those who want to alter body composition (68,69).

The Peripheral Heart Action System

The peripheral heart action system is another variation of circuit training that alternates upper body and lower body exercises throughout the circuit. This system of training distributes blood flow between the upper and lower extremities potentially improving

FIGURE 13.7

Example circuit.



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circulation. The number of exercises per sequence varies with the program's goal. The individual performs 8 to 20 repetitions per exercise, depending on the desired adaptation and phase of training they are using in the OPT model. This system is very beneficial for incorporating an integrated, multidimensional program and for altering body composition (38,70). An example for each of the three main adaptations is shown in Table 13.4.

The Split-Routine System

A split-routine system involves breaking the body up into parts to be trained on separate days. Many bodybuilders and mass-dominant and strength athletes (football, shot put, etc.) use the split-routine system. Bodybuilders typically perform numerous exercises on the same day for the same body part to bring about optimal muscular hypertrophy. By breaking up the body into parts that can be trained on different days, more work can be performed for the allotted time per workout. Examples of several different split-routines are shown in Table 13.5. Any variation of these outlined routines can be used, but the important issue with some of these routines is recovery time. When training each body part more than once a week, volume and intensity should be accounted for.

TABLE 13.4 Peripheral Heart Action System: Sample Workout

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Set 1: Stabilization	Set 2: Strength	Set 3: Power
1. Ball dumbbell chest press	1. Bench press	1. Medicine ball chest pass
2. Ball squat	2. Barbell squat	2. Squat jump
3. Single-leg cable row	3. Seated row	3. Soccer throw
4. Step-up to balance	4. Romanian deadlift	4. Power step-up
5. Single-leg dumbbell shoulder press	5. Seated dumbbell shoulder press	5. Front medicine ball oblique throw

TABLE 13.5 Split-Routine System: Sample Workouts

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Routine	Day(s) Performed	Body Parts Trained
2-Day	Monday	Chest, shoulders, triceps
	Thursday	Back, biceps, legs
3-Day	Monday	Chest, shoulders, triceps
	Wednesday	Legs
	Friday	Back, biceps
4-Day	Monday and Thursday	Chest, shoulders, triceps
	Tuesday and Friday	Back, biceps, legs

(*Continued*)

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5-Day	Monday	Chest
	Tuesday	Legs
	Wednesday	Back
	Thursday	Shoulders
	Friday	Arms
6-Day	Monday and Friday	Chest, shoulders, triceps
	Tuesday and Saturday	Legs
	Wednesday and Sunday	Back, biceps

Vertical Loading and Horizontal Loading

Vertical loading
Alternating body parts trained from set to set, starting from the upper extremity and moving to the lower extremity.

Vertical loading is a resistance training system used by NASM and follows the OPT model. It progresses a workout vertically down the template by alternating body parts trained from set to set Table 13.6. When looking at the OPT template, it can be seen that the resistance training section involves the following exercises:

1. Total body exercise
2. Chest
3. Back
4. Shoulders
5. Biceps
6. Triceps
7. Legs

In a vertically loaded workout, the client would perform the first exercise (total body) for the required repetitions and then move to the chest exercise for the next set of repetitions. After the chest exercise, the client would move on to the back exercise and so forth, until all exercises have been completed. Once completed, the client would then start back at the first exercise (total body) and run through the exercises again for the desired amount of sets. This can also be done in a circuit style, by minimizing the rest periods between exercises.

This vertical loading system of training can be very beneficial for allowing maximal recovery to each body part while minimizing the amount of time wasted on rest. For example, if it takes 1 minute to perform each exercise, by the time the client returns to the chest exercise, 7 to 10 minutes could have passed, which should be sufficient time to allow for full adenosine triphosphate (ATP)/phosphocreatine (PC) recovery. Even though 7 to 10 minutes have passed, the client has been constantly moving and has performed one set of every exercise in his or her workout.

Horizontal loading
Performing all sets of an exercise or body part before moving on to the next exercise or body part.

Horizontal loading refers to performing all sets of an exercise or body part before moving on to the next exercise or body part Table 13.7. For example, if performing three sets of a chest exercise and three sets of a back exercise, the client would perform all three sets of the chest exercise before moving on to the back exercise. The progression of exercises is therefore said to be horizontal across the template. This is the method most commonly used in health club environments and is appropriate for maximal strength and power training when longer rest periods are required between sets (71–74).

The drawback to the horizontal loading system is the amount of time typically spent resting, which can often be more time than the actual workout itself. Horizontal

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loading can be a metabolic progression if rest periods are monitored and limited to 30 to 90 seconds between sets. If the same muscle groups are forced to work with minimal recovery it can lead to faster development of metabolic and hypertrophy-related adaptations in the muscle (14,75,76).

TABLE 13.6 Vertical Loading

Exercise	Sets	Reps	Tempo	Rest	Coaching Tips
Total Body	↓				Complete exercises for each body part before beginning next set
Chest					
Back					
Shoulders					
Biceps					
Triceps					
Legs					

TABLE 13.7 Horizontal Loading

RESISTANCE					
Exercise	**Sets**	**Reps**	**Tempo**	**Rest**	**Coaching Tip**
Total Body	→				Complete all sets before moving to next exercise
Chest					
Back					
Shoulders					
Biceps					
Triceps					
Legs					

SUMMARY

The OPT method follows a progressive, systematic approach that enables personal trainers to make consistent gains with all clients by manipulating key variables to achieve various goals. There are numerous training systems that can be used to structure a resistance training program for a variety of effects.

The single-set system consists of performing one set of each exercise, usually of 8 to 12 repetitions. This system has been shown to be beneficial for strength and hypertrophy gains in beginning clients. The multiple-set system of training consists of performing a multiple number of sets for each exercise, with resistance, sets, and repetitions adjusted according to the goals and needs of the client. This system is superior to single-set training for more advanced clients.

The pyramid system involves a progressive or regressive step approach that either increases weight with each set or decreases weight with each set. The superset system uses two exercises performed in rapid succession of one another. This system is popular among bodybuilders for muscular hypertrophy and muscular endurance.

The circuit-training system programs consist of a series of exercises that an individual performs one immediately after the other, with minimal rest. This is a good system for clients with limited time and those who want to alter body composition. The peripheral heart action system is another variation of circuit training that alternates upper body and lower body exercises (of varied numbers) throughout the circuit. A split-routine system involves breaking the body up into parts to be trained on separate days so that more work can be performed for the allotted time per workout. When training each body part more than once a week, recovery period, volume, and intensity should be accounted for.

Vertical loading and horizontal loading progress a workout vertically or horizontally down the template. In a vertically loaded workout, the client would perform each exercise until all exercises have been completed and then run through the exercises again for the desired amount of sets. This can also be done in a circuit style, by minimizing the rest periods between exercises to allow maximal recovery to each body part and minimize the amount of time wasted on rest. In a horizontally loaded workout, the client would perform all sets of an exercise or body part before moving on to the next exercise or body part.

RESISTANCE TRAINING EXERCISES

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Total Body Exercise Descriptions

Total Body–Stabilization Exercises

Ball Squat, Curl to Press



Preparation

1. Begin with both feet shoulders-width apart pointing straight ahead and knees over the second and third toes. Position the ball in the low-back region as shown.

Movement

2. Lower into a squat position, keeping lower extremities in proper alignment.
3. Stand to a fully upright position while contracting the gluteal muscles and quadriceps.
4. Once stabilized, curl and press the dumbbells overhead until both arms are fully extended.
5. Slowly return to the start position and repeat.
6. Regression
   a. Decrease range of motion
7. Progression
   a. Alternating-arm
   b. One-arm
   c. Single-leg

TECHNIQUE

When performing any form of a ball squat, try to use the ball to *guide* one through the squatting motion (sitting into a chair) versus relying on the ball for support (leaning back on the ball).

RESISTANCE TRAINING EXERCISES continued

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Multiplanar Step-Up Balance, Curl, to Overhead Press



Preparation

1. Stand in front of a box with feet shoulders-width apart.

Movement

2. Step onto box with one leg, keeping foot and knee pointed straight ahead.
3. Push through heel and stand up straight, balancing on one leg.
4. Flex the other leg at the hip and knee.
5. Once balance has been established, curl and press the dumbbells overhead until both arms are fully extended.
6. Slowly return the dumbbells to the starting position.
7. Return opposite leg to the ground and step off the box.
8. Repeat on other leg.
9. Regression
   a. Omit balance
10. Progression
    a. Frontal plane
    b. Transverse plane

TECHNIQUE

When pressing overhead, make sure the low back does not arch. This may indicate tightness of the latissimus dorsi and weakness of the intrinsic core stabilizers.

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RESISTANCE TRAINING EXERCISES continued

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Total Body–Strength Exercises

Lunge to Two-Arm Dumbbell Press



Preparation

1. Begin with both feet shoulders-width apart.
2. Hold two dumbbells in hands at chest level (palms facing body).

Movement

3. Lunge forward, come to a stabilized position with front foot pointing straight ahead and knee directly over second and third toes.
4. Both knees should be bent at a 90-degree angle, front foot should be flat on the ground, and back foot should have the heel lifted off the ground.
5. From this position, drive off of front foot and back into a standing position.
6. Press the dumbbells overhead until arms are fully extended.
7. Lower weight and repeat.

TECHNIQUE

When performing any squatting or lunging motion, make sure the foot stays straight and the knees stay in line with the toes. This ensures proper joint mechanics (arthrokinematics) and optimal force generation (via proper length-tension relationships and force-couple relationships), increasing the benefit of the exercise and decreasing its risk.

RESISTANCE TRAINING EXERCISES

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Squat, Curl, to Two-Arm Press



Preparation

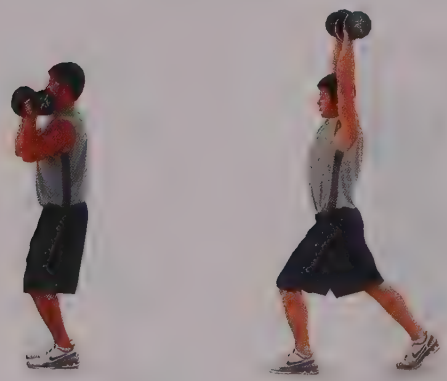
1. Begin with both feet shoulders-width apart and pointing straight ahead, and knees over the second and third toes.

Movement

2. Perform a squat as low as can be safely controlled, keeping lower extremities in proper alignment.
3. Before any compensation occurs, activate gluteal muscles and stand to a fully upright position.
4. Once stabilized, curl and press the dumbbells overhead until both arms are fully extended, with palms facing away.
5. Slowly return the dumbbells to the start position and repeat.

Total Body–Power Exercises

Two-Arm Push Press



Preparation

1. Stand with feet shoulders-width apart.
2. Hold two dumbbells in hands at shoulder level.

Movement

3. Quickly drive dumbbells up, as if doing a shoulder press.
4. At the same time, drive the legs into a stagger stance position. Back leg should be in triple extension (plantar flexion, knee extension, hip extension) with the front leg bent slightly.
5. Maintain optimal alignment on the return to the starting position and repeat.

SAFETY

One must establish proper stability (stabilization level training) and prime mover strength (strength level training) before progressing to the power exercises.

RESISTANCE TRAINING EXERCISES

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Barbell Clean



Preparation

1. Squat down next to bar, keeping feet flat on floor.
2. Arms grasp the bar fully extended slightly wider than shoulders-width apart.
3. Position shoulders over or slightly ahead of the bar, and the pelvis should be in a neutral position.

Movement

4. Initiate the first pull by extending knees and hips, lifting bar straight up (keeping the bar close to the shins) with elbows extended.
5. Continue to move bar upward by explosively extending hips, knees, and ankles (plantarflexion) in a jumping motion.
6. At maximal plantarflexion, elevate shoulders and flex and pull with the arms.
7. Pull bar as high as possible before rotating elbows under the bar.
8. Hyperextend wrists as the elbows move under the bar.
9. Point elbows forward.
10. Rack the bar across the front of the shoulders, keeping torso erect.
11. Flex hips and knees to absorb the weight of the bar.

SAFETY

This is a simplified description and illustration of performing the barbell clean. The barbell clean is an advanced power exercise and requires proper instruction before attempting.

RESISTANCE TRAINING EXERCISES continued

Chest Exercise Descriptions

Chest-Stabilization Exercises

Ball Dumbbell Chest Press



Preparation

1. Lie on stability ball as shown.
2. Maintain a bridge position by contracting gluteal muscles and keeping shoulders, hips, and knees at the same level.
3. Feet should be shoulders-width apart with toes pointing straight ahead.
4. Hold one dumbbell in each hand, at chest level.

Movement

5. Press both dumbbells straight up and then together, by extending elbows and contracting chest.
6. Hold.
7. Slowly return dumbbells toward body by flexing elbows.
8. Regression
   a. Dumbbell chest press progression on bench
9. Progressions
   a. Alternating-arm
   b. Single-arm

SAFETY To ensure proper alignment, the ears, shoulders, hips, and knees should all be in line with one another.

RESISTANCE TRAINING EXERCISES continued

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Push-Up



Preparation

1. Begin in a push-up position as shown.
2. Draw in navel and contract gluteal muscles.

Movement

3. Keeping pelvis in a neutral position, slowly lower body toward ground, by flexing elbows and retracting and depressing shoulder blades.
4. Push back up to starting position, by extending elbows and contracting chest. Do not allow head to jut forward.
5. Regressions
   a. On knees b. Hands on bench, feet on floor
   c. Hands on wall, feet on floor
6. Progressions
   a. Lower extremities on ball b. Hands on medicine balls c. Hands on stability ball

SAFETY

A common compensation that occurs when performing a push-up is the low back arching (stomach falls toward the ground). This is an indicator that the individual possesses weak intrinsic core stabilizers, and the exercise must be regressed.

Chest-Strength Exercises

Flat Dumbbell Chest Press



TECHNIQUE

When performing chest presses, the range of motion at the shoulder joint (how far the elbows go down) will be determined by the load one is lifting (control) and tissue extensibility. The key is to only go as far as one can control without compensating.

Preparation

1. Lie on flat bench with knees bent and feet flat on the floor.
2. Hold one dumbbell in each hand, at chest level.

Movement

3. Press both dumbbells straight up and then together, by extending elbows and contracting chest.
4. Hold.
5. Slowly return dumbbells toward body by flexing elbows and allowing shoulders to retract and depress.

RESISTANCE TRAINING EXERCISES continued

Barbell Bench Press



Preparation

1. Lie on flat bench, feet flat on floor and toes pointing straight ahead.
2. Hold a barbell and grasp the bar with hands slightly wider than shoulders-width apart.

Movement

3. Slowly lower the bar toward the chest by flexing elbows. Avoid letting the back arch or the head jut off the bench.
4. Press the bar back up, extending arms and contracting chest, until elbows are fully extended.

Chest-Power Exercises

Two-Arm Medicine Ball Chest Pass



Preparation

1. Stand facing a wall or partner and establish a squared stance position.
2. Hold a medicine ball (5 to 10% of body weight) with both hands, elbows flexed, at chest level

Movement

3. Push and release the ball straight ahead as hard as possible, by extending the elbows and contracting the chest. Do not allow the shoulders to elevate.
4. Catch and repeat as quickly as possible..

TECHNIQUE

If it is not an option to be able to perform power exercises with a medicine ball (facility-dependent, equipment), this exercise can also be done using tubing or cable. Just make sure to adjust the weight or resistance accordingly so one can still perform the movement quickly and under control without compensation.

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RESISTANCE TRAINING EXERCISES

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Rotation Chest Pass



Preparation

1. Stand, with body turned at a 90-degree angle from a wall or partner.
2. Hold a medicine ball (5 to 10% of body weight) with both hands, elbows flexed, at chest level.

Movement

3. Use abdominal muscles, hips, and gluteal muscles to rotate body quickly and explosively toward the wall.
4. As body turns, pivot back leg and allow it to go into triple extension (plantarflexion, knee extension, hip extension).
5. With the upper body, push the medicine ball using the back arm to extend and apply force as shown.
6. Catch and repeat as quickly as possible, under control.

RESISTANCE TRAINING EXERCISES continued

Back Exercise Descriptions

Back-Stabilization Exercises

Standing Cable Row



Preparation

1. Stand facing a cable machine with feet pointing straight ahead.
2. Hold cables, with arms extended at chest level.

Movement

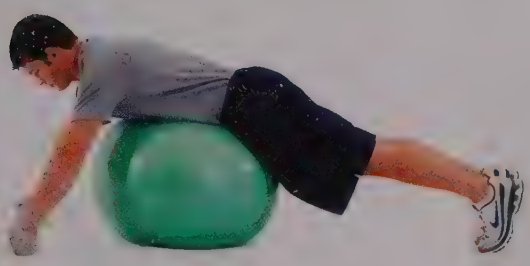
3. With knees slightly flexed, row cable by flexing elbows, retracting and depressing the shoulder blades.
4. Do not allow the head to jut forward or shoulders to elevate.
5. Hold.
6. Slowly return arms to original position, by extending the elbows.
7. Regression
   a. Seated
8. Progressions
   a. Two-legs, alternating-arm
   b. Two-legs, one-arm
   c. Single-leg, two-arms
   d. Single-leg, alternating-arm
   e. Single-leg, one-arm

TECHNIQUE

When performing rows, initiate the movement by retracting and depressing the shoulder blades (scapulae). Do not allow the shoulders to elevate.

RESISTANCE TRAINING EXERCISES continued

Ball Dumbbell Row





Preparation

1. Begin in a prone position, with stability ball under abdomen.
2. Keep feet pointed down, legs completely straight.
3. Hold dumbbells in each hand and extend arms in front of body.

Movement

4. Contract gluteal muscles and quadriceps.
5. Row the dumbbells by flexing elbows and retracting and depressing shoulder blades.
6. Hold.
7. Return dumbbells slowly to ground, by extending elbows.
8. Regression
   a. Kneeling over ball
9. Progression
   a. Alternating-arm
   b. One-arm

TECHNIQUE

Performing exercises in a prone position can be uncomfortable. When working with overweight individuals, it may be more appropriate to perform these exercises in a seated or standing position.

RESISTANCE TRAINING EXERCISES continued

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Back-Strength Exercises

Seated Cable Row



Preparation

1. Sit facing a cable machine, with feet shoulders-width apart and pointing straight ahead.
2. Hold cables, with arms extended.

Movement

3. Row cable by flexing elbows and pulling the handles toward the trunk.
4. Bring thumbs toward the armpits, keeping the shoulder blades retracted and depressed.
5. Do not allow the head to jut forward.
6. Hold.
7. Slowly return arms to original position by extending the elbows.

TECHNIQUE

To increase the effectiveness of the exercise and decrease the risk of injury, keep the torso stationary throughout the execution of the exercise; flexing and extending the torso while performing the row creates momentum, which decreases the effectiveness of the exercise and places stress on the low back.

RESISTANCE TRAINING EXERCISES continued

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Seated Lat Pulldown





Preparation

1. Sit upright with feet on the floor pointing straight ahead.

Movement

2. Pull handles toward the body, by flexing elbows and depressing the shoulder blades. Do not arch back, allow head to jut forward, or elevate the shoulders.
3. Hold at end range.
4. Slowly return weight to original position, by extending elbows.

SAFETY

Performing lat pulldowns with a bar behind the neck is not advised as this places stress to the shoulder joints and cervical spine. If performing the lat pulldown exercise with a bar (instead of cables as illustrated), the bar should pass in front of the face approximately to shoulder height.

RESISTANCE TRAINING EXERCISES continued

Back-Power Exercises

Medicine Ball Pullover Throw



Preparation

1. Place stability ball under low back, bend knees at a 90-degree angle, and keep feet flat and toes pointing straight ahead.
2. Hold a medicine ball (5 to 10% of body weight) overhead with both hands, with arms extended.

Movement

3. Using abdominals, quickly crunch forward.
4. Throw the medicine ball straight ahead keeping elbows fairly straight.
5. As the ball releases, continue pulling the arms through all the way to the sides of the body.
6. Keep chin tucked throughout the exercise.
7. Repeat as quickly as possible, under control.

SAFETY

To decrease stress to the shoulder and low back, it will be important that one has optimal extensibility through the latissimus dorsi musculature before performing these back-power exercises.

Soccer Throw



Preparation

1. Stand holding a medicine ball (5 to 10% of body weight) overhead as illustrated.

Movement

2. Quickly throw the medicine ball toward the floor, allowing the arms to follow through.
3. Repeat.

RESISTANCE TRAINING EXERCISES continued

Shoulder Exercise Descriptions

Shoulder-Stabilization Exercises

Single-Leg Dumbbell Scaption



Preparation

1. Hold dumbbells at side, with palms facing side of body.
2. Raise one foot off the floor as illustrated.

Movement

3. Raise both arms, thumbs up, at a 45-degree angle in front of the body, until hands reach approximately eye level.
4. Keep shoulder blades retracted and depressed throughout the exercise. Do not allow the back to arch.
5. Hold.
6. Slowly return arms back to sides of body and repeat.
7. Regression
   a. Two-legs
   b. Seated
8. Progression
   a. Single-leg, alternating-arm
   b. Single-leg, single-arm
   c. Proprioceptive modalities

TECHNIQUE

Performing shoulder exercises in the scapular plane decreases the risk of the supraspinatus muscle becoming impinged between the head of the humerus and the coracoacromial arch of the scapula.

RESISTANCE TRAINING EXERCISES continued

Seated Stability Ball Military Press



Preparation

1. Sit on a stability ball, keeping the toes pointed straight ahead, feet hips-width apart.
2. Hold dumbbells at shoulder level.

Movement

3. Press the dumbbells overhead until both arms are fully extended, with palms facing away.
4. Hold.
5. Slowly return dumbbells to the start position and repeat.
6. Regression
   a. Seated on a bench
7. Progression
   a. Alternating-arm
   b. One-arm
   c. Standing

SAFETY

Performing exercises on a stability ball can be uncomfortable for some people. It may be required for the health and fitness professional to hold the ball while the individual performs the exercise to provide some additional support (both mentally and physically).

RESISTANCE TRAINING EXERCISES *continued*

Shoulder-Strength Exercises

Seated Dumbbell Shoulder Press



Preparation

1. Sit on a bench, feet flat on floor, and toes pointing straight ahead.
2. Hold dumbbells in each hand as illustrated.

Movement

3. Press arms directly overhead by extending elbows. Do not allow head to jut forward or back to arch.
4. Slowly return to starting position and repeat.

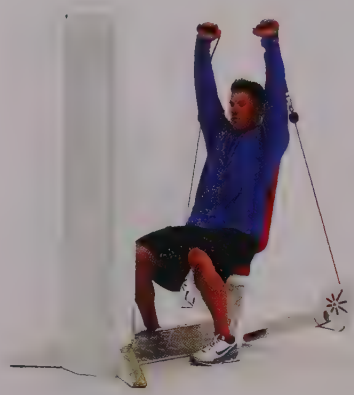
SAFETY

When performing overhead presses, make sure the cervical spine stays neutral (head drawn back). Do not allow the head to migrate forward as this places excessive stress on the posterior neck muscles and cervical spine.

RESISTANCE TRAINING EXERCISES continued

Seated Shoulder Press Machine





Preparation

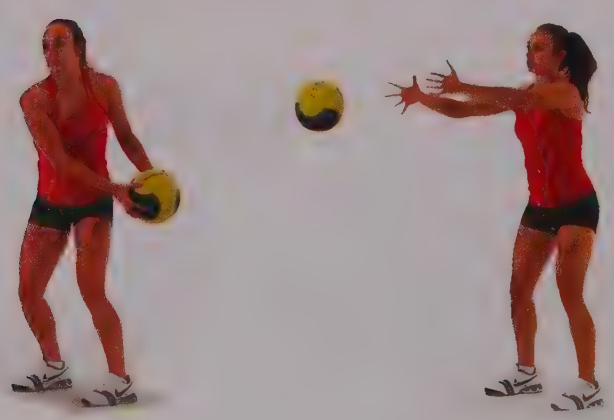
1. Sit at machine with feet straight ahead.
2. Make any adjustments necessary to fit body.
3. Select desired weight.
4. Keep the chin tucked.

Movement

5. Press weight overhead until arms are fully extended.
6. Hold.
7. Slowly return weight to original position by flexing elbows and allowing shoulder blades to retract and depress.

Shoulder-Power Exercises

Front Medicine Ball Oblique Throw



Preparation

1. Stand facing a wall or partner with feet shoulders-width apart, knees slightly bent, and toes pointing straight ahead.
2. Hold a medicine ball (between 5 and 10% of body weight) as illustrated.

Movement

3. Toss the ball against the wall or to a partner with an underhand motion.
4. Use a scooping motion to catch ball.
5. Repeat as quickly as can be controlled.
6. This exercise can be performed continuously to one side or by alternating sides.

TECHNIQUE

If a partner is unavailable, you can perform the exercise by tossing the medicine ball against a wall.

RESISTANCE TRAINING EXERCISES (continued)

Overhead Medicine Ball Throw



Preparation

1. Stand with back to wall in a semi-squat position holding a medicine ball as shown.

Movement

2. Jump off the ground explosively, extending arms overhead.
3. Throw the medicine ball off the wall.
4. Release the ball before arms pass ears. Do not allow the back to go into hyperextension.
5. Land in a controlled and stable manner, avoiding all compensations.

Biceps Exercise Descriptions

Biceps–Stabilization Exercises

Single-Leg Dumbbell Curl



Preparation

1. Stand on one foot and allow arms to extend and hang to the sides of the body with a dumbbell in each hand.

Movement

2. Perform a biceps curl by performing elbow flexion and supination.
3. Slowly return dumbbells to their original position.
4. Regression
   a. Two-leg
5. Progression
   a. Alternating-arm
   b. Single-arm
   c. Proprioceptive modalities

TECHNIQUE

Keeping the scapulae retracted during the exercise ensures proper scapular stability, placing more of an emphasis on the biceps musculature.

RESISTANCE TRAINING EXERCISES

Single-Leg Barbell Curl



Preparation

1. Stand on one foot while holding a barbell in both hands as shown.

Movement

2. Perform a barbell curl by flexing both elbows, keeping the shoulder blades retracted.
3. Curl bar up to chest level.
4. Slowly lower the bar back to original position by extending the elbows.
5. Regression
   a. Two-leg
6. Progression
   a. Proprioceptive modalities

SAFETY

To decrease stress on the elbow, do not grip too close or too wide on the bar. To determine grip width, extend your elbows so your hands fall naturally to your sides, palms facing forward. Where your hands fall at your sides is the position where they should be when they grip the bar.

Biceps–Strength Exercises

Seated Two-Arm Dumbbell Biceps Curl



Preparation

1. Sit on a bench with feet flat on the floor, pointing straight ahead.
2. Hold dumbbell in each hand with arms at sides.

Movement

3. Perform a biceps curl by performing elbow flexion and supination.
4. Keep shoulder blades retracted throughout the exercise.
5. Slowly return dumbbells to their original position.

RESISTANCE TRAINING EXERCISES continued

Biceps Curl Machine



Preparation

1. Sit at machine.
2. Make any adjustments necessary to fit body.
3. Select desired weight.

Movement

4. Perform a curl by flexing elbows, while keeping shoulder blades retracted. Do not allow head to jut forward.
5. Curl until end range.
6. Slowly return weight to original position by extending elbows.

TECHNIQUE

When performing biceps curls it is important to keep an upright posture. Do not allow your torso to excessively flex or extend to cheat the movement.

RESISTANCE TRAINING EXERCISES continued

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Triceps Exercise Descriptions

Triceps–Stabilization Exercises

Supine Ball Dumbbell Triceps Extensions



Preparation

1. Lie on a stability ball, with ball between shoulder blades.
2. Maintain a bridge position by contracting gluteal muscles and keeping shoulders, hips, and knees level.
3. Feet should be shoulders-width apart with toes pointing straight ahead.

Movement

4. Extend elbows until arms are straight.
5. Return dumbbells slowly to the start position by flexing elbows.
6. Regression
   a. On bench
7. Progressions
   a. Alternating-arms
   b. One-arm

SAFETY

When performing stability ball exercises in a supine position, make sure position is such that head comfortably rests on the ball. This will decrease stress to the cervical spine.

Prone Ball Dumbbell Triceps Extensions



Preparation

1. Lie in a prone position, with stability ball under abdomen.
2. Keep feet pointed down, legs completely straight, and navel drawn in.
3. Hold dumbbells in each hand with elbows bent and shoulder blades retracted and depressed.

Movement

4. Maintain a retracted scapular position, and extend elbows so that they are parallel to the side of the body.
5. Hold.
6. Slowly return dumbbells to original position by flexing elbows.
7. Regression
   a. Standing with cable
8. Progression
   a. Alternating-arms
   b. Single-arm

TECHNIQUE

To ensure optimal alignment, make sure the ankles, knees, hips, elbows, shoulders, and ears are all in alignment and maintained throughout the exercise.

RESISTANCE TRAINING EXERCISES continued

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Triceps-Strength Exercises

Cable Pushdown



Preparation

1. Stand with feet shoulders-width apart, pointed straight ahead, and knees slightly flexed.
2. Grasp a cable, with elbows flexed.
3. Keep shoulder blades retracted and depressed.

Movement

4. Contract triceps by pushing hands toward the ground until arms are fully extended.
5. Hold.
6. Slowly return to the starting position.
7. Repeat.

TECHNIQUE

Using a rope when performing cable pushdowns will allow the elbows to track through their natural path of motion versus having the hands closely fixed on a bar. This may help decrease the risk of compensation when performing the exercise.

Supine Bench Barbell Triceps Extension



SAFETY

As with barbell curls, keeping the hands too close on the bar can increase stress on the elbow. Having your hands closer to shoulders-width apart can help to decrease stress to the elbow and compensation.

Preparation

1. Lie on flat bench.
2. Feet should be flat with toes pointing straight ahead.
3. Hold barbell in both hands with elbows flexed as illustrated.

Movement

4. Extend elbows until arms are straight.
5. Hold.
6. Slowly lower barbell toward forehead by flexing the elbows.
7. Repeat.

RESISTANCE TRAINING EXERCISES continued

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Leg Exercise Descriptions

Leg-Stabilization Exercises

Ball Squat





SAFETY

Ball squats are a great way to teach individuals how to squat properly, with the goal to have them eventually progress to squats without the stability ball.

Preparation

1. Stand with feet shoulders-width apart, toes pointing forward, and knees over second and third toes. Hold dumbbells to the side of the body.
2. Rest back against a stability ball, which is placed on a wall.
3. Ideally, keep feet under the knees. For individuals who lack ankle dorsiflexion (tight calves), place the feet slightly in front of the knees.

Movement

4. Slowly begin to squat down, bending knees and flexing hips, keeping feet straight (as if sitting into a chair). Keep the knees in line with the toes and abdominals activated.
5. Allow the pelvis to sit back under the ball while maintaining a neutral spine.
6. Keep the chest up and put pressure through the heels. Do not rely solely on the ball for support.
7. To rise back up, contract gluteal muscles and place pressure through the heels as knees are extended.
8. Stand up straight until hips and legs are fully extended. Avoid compensation in the low back or lower extremities.
9. Regression
   a. Decrease range of motion
   b. Holding on to a stable support
10. Progression
    a. Squat without stability ball

RESISTANCE TRAINING EXERCISES continued

Multiplanar Step-Up to Balance



Preparation

1. Stand in front of box, with feet pointed straight ahead. Hold dumbbells to the side.

Movement

2. Step onto box with one leg, keeping foot pointed straight ahead and knee lined up over mid-foot.
3. Push through heel and stand up straight, balancing on one leg.
4. Flex the other leg at the hip and knee and dorsiflex the foot.
5. Return "floating" leg to the ground and step off the box, maintaining optimal alignment.
6. Repeat on other leg.
7. Regression
   a. Omit balance
   b. Decrease step height
8. Progression
   a. Frontal plane step-up
   b. Transverse plane step-up

TECHNIQUE

Lunges are excellent lower-extremity strengthening exercises; however, many individuals lack the flexibility and stabilization requirements to execute the exercise properly. Step-ups are a great way to regress the lunge until one develops proper flexibility and stabilization capabilities to perform the lunge correctly.

RESISTANCE TRAINING EXERCISES continued

Leg-Strength Exercises

Leg Press (Hip Sled)



TECHNIQUE

Make sure the feet are positioned on the platform hips- to shoulders-width apart, toes are pointed straight, and the knees track in line with the toes. This will decrease stress to the knees, hips, and low back.

Preparation

1. Stand in machine.
2. Make any adjustments necessary to fit body.
3. Place feet shoulders-width apart, with toes pointing forward and knees directly over second and third toes.

Movement

4. Slowly perform a squat to a depth that can be safely controlled. Keep optimal alignment of the lower extremities throughout the movement.
5. Before any compensation occurs, activate gluteal muscles and apply pressure through heels extending hips and knees back to the starting position.

RESISTANCE TRAINING EXERCISES (continued)

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Barbell Squat



Preparation

1. Stand with feet shoulders-width apart, toes pointing straight ahead, and knees over second and third toes.
2. Rest barbell on shoulders, behind neck, with hands grasping the bar wider than shoulders-width apart.

Movement

3. Slowly begin to squat down, bending knees and flexing hips, keeping feet straight. Do not allow the knees to move inward.
4. Keep the chest up and put pressure through the heels.
5. Squat to a depth that can be safely controlled while maintaining ideal posture.
6. To rise back up, contract gluteal muscles and place pressure through the heels as knees and hips are extended.
7. Stand up straight until hips and legs are fully extended. Avoid compensation in the low back or lower extremities.

SAFETY

How far down should you squat? Only as far as can be controlled without compensating. As one develops more flexibility and stabilization strength, the range of motion can be increased, assuming no compensations occur.

RESISTANCE TRAINING EXERCISES continued

Leg-Power Exercises

Squat Jump



Preparation

1. Stand with feet shoulders-width apart and pointing straight ahead.

Movement

2. Squat down slightly.
3. Jump up into the air while extending arms overhead.
4. Bring arms back to sides during landing.
5. Land softly on the mid-foot in a controlled manner with feet and knees pointing straight ahead. Repeat as quickly as can be controlled.

Tuck Jump



Preparation

1. Stand with feet shoulders-width apart and pointing straight ahead.

Movement

2. Jump up off the ground and, while in the air, bring knees up to chest.
3. Land softly on the midpoint of the feet, in a controlled manner, with feet straight and knees over mid-foot.
4. Repeat, as quickly as can be controlled.

TECHNIQUE

When performing power exercises, make sure you land behind the ball of the foot (not on the ball of the foot or on the heel). This will ensure proper force distribution through the foot and lower extremity, improving force production capabilities.

Filling in the Template

The information contained in this chapter provides personal trainers with the essentials to effectively design safe and effective resistance training programs for the needs of a variety of clients. The information provided in this chapter combined with the information contained in chapter 15 allows the personal trainer to properly complete the resistance training portion of the OPT program template. The template should be completed according to the example in **Figure 13.8**.

Professional's Name: Brian Sutton

CLIENT'S NAME: JOHN SMITH **DATE: 5/01/13**

GOAL: TOTAL BODY **PHASE: 1 STABILIZATION ENDURANCE**

WARM-UP

Exercise:	Sets	Duration	Coaching Tip
SMR: Calves, IT-Band, Lats	1	30 s.	Hold each tender area for 30 sec
Static Stretch: Calves, Hip Flexors, Lats	1	30 s.	Hold each stretch for 30 sec
Treadmill	1	5–10 min	Brisk walk to slow jog

CORE / BALANCE / PLYOMETRIC

Exercise:	Sets	Reps	Tempo	Rest	Coaching Tip
Floor Bridge	2	12	Slow	0	
Floor Prone Cobra	2	12	Slow	0	
Single-Leg Balance Reach	2	12	Slow	0	
Squat Jump w/Stabilization	2	5	Slow	90 s.	Hold landing 3–5 seconds

SPEED, AGILITY, QUICKNESS

Exercise:	Sets	Reps	Rest	Coaching Tip
Optional				
Optional				

RESISTANCE

Exercise:		Sets	Reps	Tempo	Rest	Coaching Tip
Total Body	Ball Squat, Curl to Press	2	12	Slow	0	Vertical loading
Chest	Ball Dumbbell Chest Press	2	12	Slow	0	
Back	Standing Cable Row	2	12	Slow	0	
Shoulders	Single-Leg Dumbbell Scaption	2	12	Slow	0	
Biceps	Single-Leg Dumbbell Curl	2	12	Slow	0	
Triceps	Supine Ball Dumbbell Triceps Extensions	2	12	Slow	0	
Legs	Step-Up to Balance: Sagittal Plane	2	12	Slow	90 s.	

COOL-DOWN

Exercise:	Sets	Duration	Coaching Tip
Treadmill (optional)	1	5–10 min	Brisk walk; gradually reduce speed
SMR: Calves, IT-Band, Lats	1	30 s.	Hold each tender area for 30 sec
Static Stretch: Calves, Hip Flexors, Lats	1	30 s.	Hold each stretch for 30 sec

Coaching Tips:

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FIGURE 13.8 Completed resistance training section of the template.

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